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Department of Computer Science and Engineering (AI&ML)

Major ProjectReview-1(Semester VII)

Hierarchical Position Encoding for Structure-Sensitive Semantic Dense Retrieval

Group No : 18

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Outline of Presentation

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- Introduction
- Goals and Objectives
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- Literature Review (Atleast 10 to15 Papers in Tabular Format)
- Existing System & its Limitation
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- Methodology
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- Modules /Functionalities
- Technology & Requirements
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- Project Timeline (Work Completed,currect Stage ,Upcoming Activities)
- Conclusion & Future Scope
- References

Problem Definition

- Every production embedding model encodes token position as a single integer: sequence position 1, 2, 3.., 47...
- The dominant mechanism is **Rotary Position Embedding** (RoPE), which produces attention scores reflecting only linear distance between tokens.
- This architecture is blind to structural address : the heading path, table row-column coordinate, JSON key-path that governs what a token actually means in **hierarchically structured documents**.

- The result is a systematic retrieval failure: tokens with near-identical surface text but structurally-opposed semantic roles are assigned dangerously similar embedding vectors, causing RAG systems to surface structurally wrong context and language models to reason confidently from incorrect information.

- Embedding Models which are widely used; None of them structurally discriminate - because the architectural blind spot exists at the position encoding layer. Scaling training data over a broken inductive bias produces a more confident wrong answer, not a correct one.
- The problem is not data size. It is that RoPE encodes one number per token sequence position and throws away every other dimension of where a token actually lives in a document.
- The model did not fail on vocabulary. It failed on structure. The architecture had no way to know these two sentences meant opposite things.
- When the same sentence means opposite things depending on where it appears in a document, can the embedding model tell the difference?

Introduction

- The global RAG market is projected to grow from $2.33B in 2025 to $81.51B by 2035 (CAGR 42.7%), driven primarily by document retrieval, which accounts for 32.4% of all RAG revenue. Over 80% of enterprise data being retrieved is unstructured
- 71% of organizations use GenAI regularly yet only 17% report meaningful business impact.[1] The bottleneck is not model capability. It is retrieval quality.

Document → Chunk → Embed (RoPE) → Vector DB → Query Match → LLM

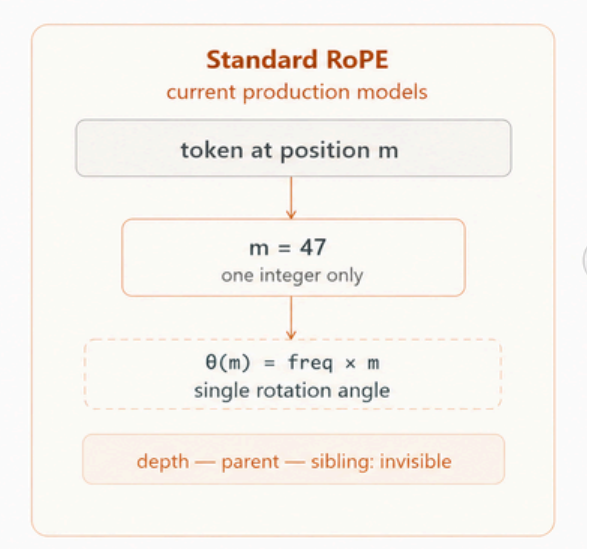
- BGE-M3, Qwen3-Embedding, E5-Mistral, Nomic Embed, all encode token position as a single integer via RoPE. Linear sequence distance is the only positional signal available to the model during embedding

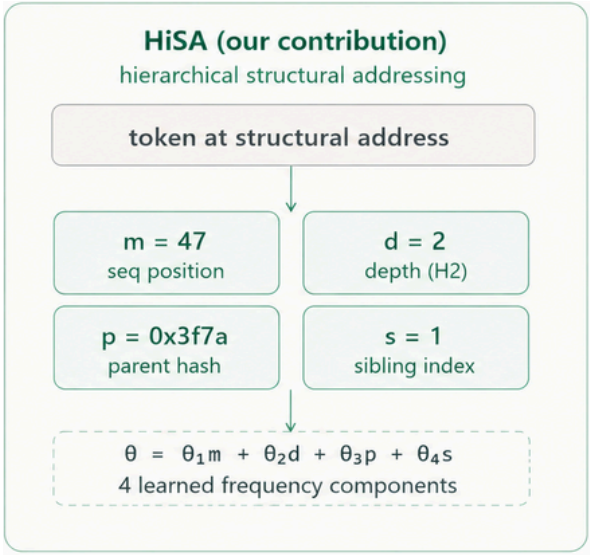
- Naive RAG achieves only 10–40% success rates in enterprise environments. Retrieval failure not generation failure is the primary cause of wrong answers in production AI systems.

- Every production RAG pipeline today is only as good as its embedding model and every embedding model has the same architectural blind spot.

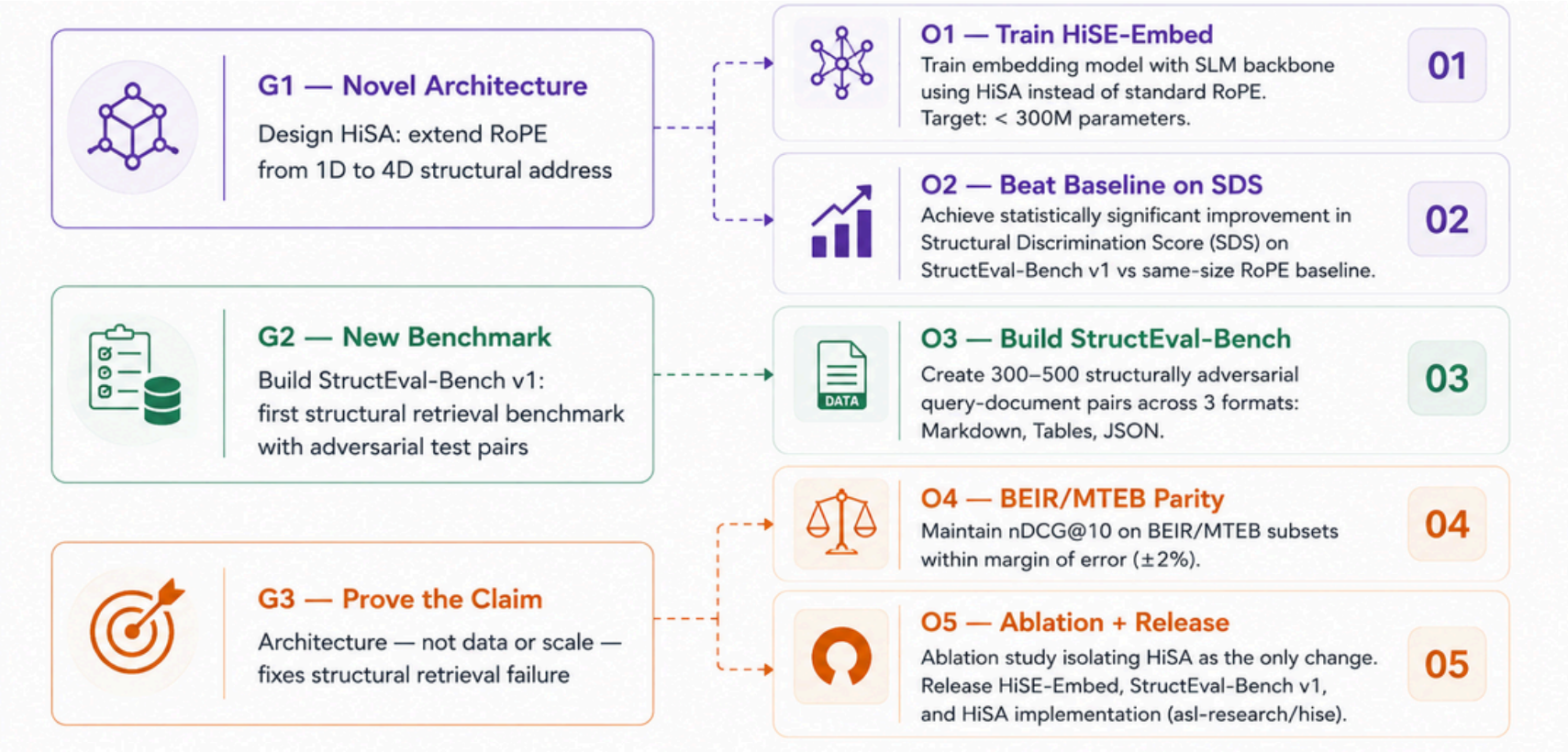
Introduction

- RoPE computes Attention(A,B) ~ f(|pos_A - pos_B|) - sequence distance only. Two tokens under Risks vs Performance are structurally opposed but positionally identical. Depth, parent path, sibling index mathematically invisible to every production model.
- TUTA, DOM-LM, TabBiN, Struc-EMB all prompt level or classification-task workarounds. None touched the positional encoding layer. BEIR and MTEB have zero structurally-adversarial test pairs - the benchmark never exposed the failure, so no model ever optimized against it.
- Enterprise RAG runs over 10K filings, legal contracts, Kubernetes configs, clinical protocols - all hierarchically structured. The higher the structural complexity, the more confidently wrong the retrieval becomes.
- **No production model encodes where a token lives in a document. HiSE fixes this at the position encoding layer - not around it.**





Goals and Objectives



- Hypothesis: Explicit structural positional encoding enables transformers to distinguish semantically similar but structurally different content.
- Validation Strategy: Verify the hypothesis through controlled benchmarking, ablation studies, and comparison with standard RoPE-based embedding models.

Motivation

Structure Carries Meaning:
Headings, tables, and key paths encode semantics beyond textual content.

Current Models Are Structure-Blind:
Modern embedding architectures preserve token order—not document hierarchy.

Retrieval Fails Before Generation:
Incorrect embeddings propagate structural errors throughout the RAG pipeline.

HiSE Addresses the Root Cause:
Model structural context explicitly instead of scaling parameters or training data.

- Research Gap in Positional Encoding: Existing positional encoding methods primarily model sequential relationships. Hierarchical structural position remains largely unexplored for dense retrieval.
- Embedding Quality Determines Retrieval: Dense retrieval systems are fundamentally limited by the quality of their embedding representations. Improving positional encoding directly improves retrieval accuracy.

- HiSE is motivated by the need to make dense retrieval understand not only what information says,but also where it exists within a document's structure

Literature Review

Paper	Year	What it did	Why it's not enough
TUTA (Microsoft)	2021	Bi-dimensional tree coords for table structure	Classification only - no retrieval training
DOM-LM	2022	Depth + parent + sibling PE in BERT for HTML	HTML-only, additive not rotational, no RAG
Tree-Enhanced CodeBERTa	2025	AST structural PE in CodeBERTa for code	Code/AST only, not a retrieval model
TabBiN (EDBT)	2025	2D tabular coords + visibility matrix	Table-specific, no cross-format, no dense retrieval
SEAL (EMNLP)	2025	Structure-aware contrastive learning	Ain`t introduce a unified rotational positional encode
Struc-EMB (Georgia Tech)	2025	Structural context in LLM-based embedders	Prompt-level workaround, not architectural PE
Rethink RoPE (IMP)	2025	N-dimensional RoPE extension blueprint	Vision/spatial only — not applied to documents
RoPE (IMP)		Encoding relative positional information through rotations	does not model hierarchical structure directly
Attention is all you need	2017	sinusoidal positional encoding for sequence	no dense retrieval objective & linear sequence pos...
Adaptive Quantitative: IDRLA		Introduce Partially observable Markov Decision Process	#

Existing System & its Limitation

Every approach tried to work around the problem. None touched where the problem lives.

System	What it does	Core Limitation
BGE-M3 (570M)	State-of-art dense retrieval, multi-lingual	RoPE-based, 1D position only. nDCG@10: 0.467. No structural signal
Nomic Embed v2 (137M)	Open-source dense retrieval	Same RoPE architecture. Cosine sim ≈ 0.97 on structurally-opposed chunks
Qwen3-Embedding-0.6B	retrieval model for QWEN source model	Linear sequence distance only. Structural info: architecturally invisible
E5-Mistral-7B	Microsoft large-scale embedder	Scaled parameters over broken inductive bias, more confident wrong answer
Struc-EMB (Georgia Tech, 2025)	Structural context via prompt concatenation	Prompt-level workaround — positional encoding layer untouched

Every existing system encodes where a token appears in a sequence. None encodes where a token lives in Semantics.

Scaling data and parameters over a broken inductive bias produces a more confident wrong answer, not a correct one.

Research Gap

No system has ever encoded where data lives in its own structure — only where it appears in a sequence.

1. **Every production model encodes position as a single integer via RoPE**. Tabular coordinates, JSON key-paths, HTML node depth, Markdown heading hierarchy all collapsed to m = 47. One number. Every structural dimension discarded.
2. **No existing work produces a composite structural address across data formats**. TUTA solved tables. DOM-LM solved HTML. Struc-EMB used prompt concatenation. None combined (m, d, p, s) into a unified rotational signal embedded into the attention mechanism itself — format-agnostic, architecture-level.
3. **RoPE operates on a 1-dimensional rotational manifold — one circular direction in attention space**. No system extends this to a multi-dimensional rotational geometry where depth, parent identity, and sibling index each contribute independent learned frequencies, making structural proximity a geometric property — not just a training outcome.
4. **No system applies POMDP to structural address inference**. Full document parse at token level is computationally intractable at scale. A Partially Observable Markov Decision Process over structural state gives high-probability composite address estimation — processing just enough structure to achieve true-negative precision without exhaustive tree computation.

Three prospects : structural understanding, data encoding, dense retrieval each solved a piece. None converged. The composite structural address across all data formats, embedded into the rotational geometry of attention itself, with probabilistic structural inference via POMDP, has never been built.

Proposed System

HiSA- Hierarchical Structural Addressing · From Single Integer to Composite Structural Address

1. Every data format- tabular, JSON, HTML/XML, Markdown, complex documents shares one failure: position encoded as a single integer m. HiSA replaces this with a format-agnostic n-dimensional structural address (m, d, p, s): sequence position, tree depth, parent section identity, and sibling index extracted by format-specific parsers, fed into a single unified encoding mechanism.
2. Each dimension gets its own learned rotation frequency. The composite structural address is not concatenated — it is embedded into the rotational geometry of attention itself. Two tokens under the same structural node stay geometrically proximate even 200 sequence positions apart. Structurally-opposed tokens are pushed apart by the geometry — not just by training signal.
3. The core formula changes from $\theta = \theta_1 \times m$ (one rotation, one manifold) to $\theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$ (four rotations, 4-dimensional manifold). Only 96 new scalar parameters added — surgically small change, structurally large effect.
4. POMDP over structural state makes this computationally viable. Full structural parse of every token at scale is intractable. A Partially Observable Markov Decision Process estimates the composite structural address probabilistically — processing just enough document structure to achieve true-negative level precision without exhaustive tree computation at every token.
5. Degenerate case guarantee — when data has no structure, d=0, p=0, s=0, and HiSA reduces exactly to standard RoPE. No regression on plain-text retrieval benchmarks like BEIR and MTEB.

Same backbone · same data · same compute · only positional encoding changes improvement isolates HiSA as the experimental variable

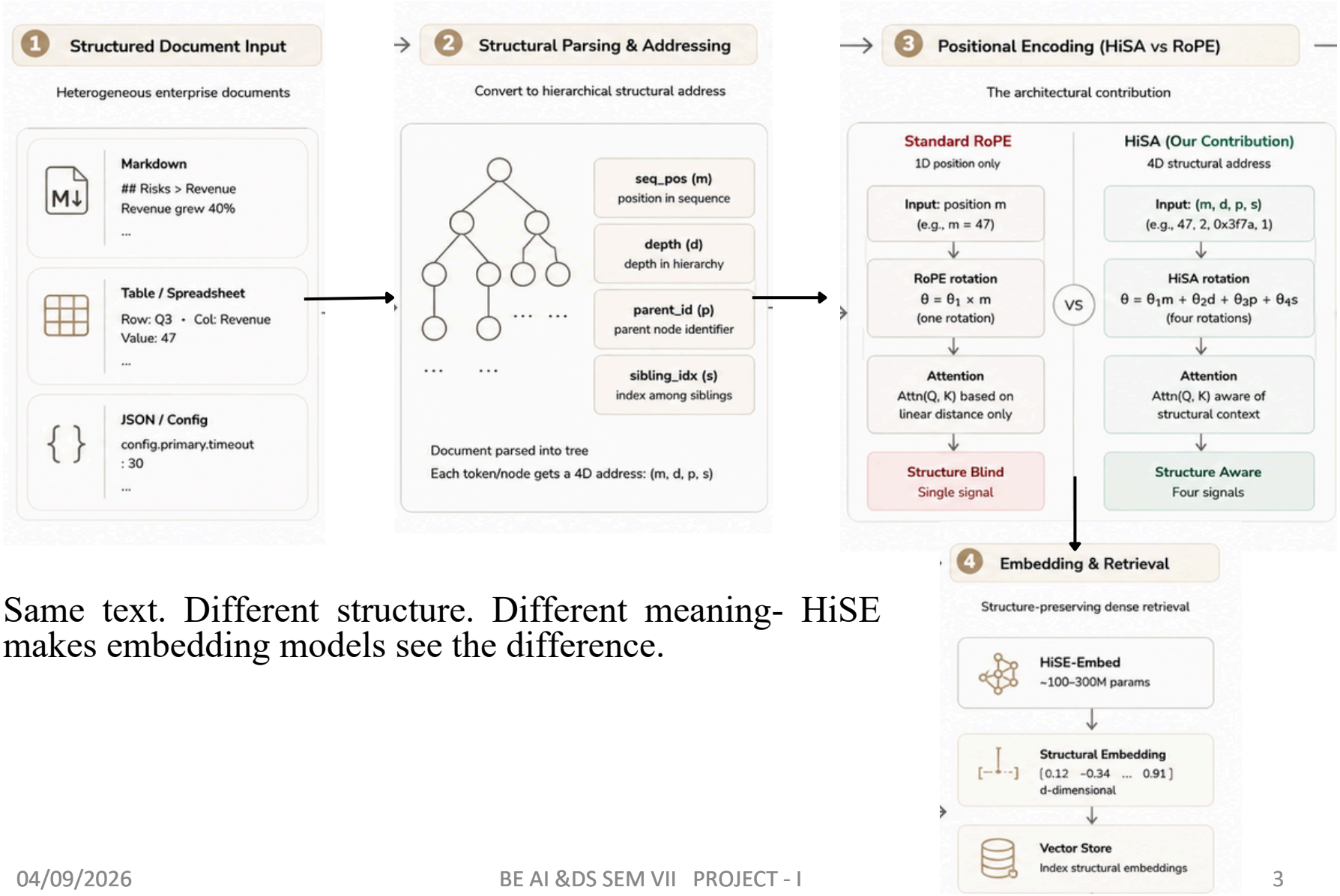
Methodology

Five-phase experimental design — architecture first, benchmark second, proof third.

1. **Failure-Mode Survey :** Quantitatively prove the structural discrimination gap. Run BGE-M3, Qwen3-Embedding, Nomic Embed on structurally-adversarial pairs across tabular, JSON, HTML, Markdown. Establish baseline cosine similarity scores and Retrieval Inversion Rate (RIR). Target: cosine sim ≥ 0.90 on structurally-opposed pairs.
2. **HiSA Implementation** : Replace standard RoPE with 4D structural address in a small SLM backbone. Build format-specific parsers for all data types — Markdown AST, pandas table extractor, JSON recursive key-path, HTML/XML DOM walker — all outputting unified (m, d, p, s) tuples. Unit test: d=0, p=0, s=0 produces identical output to standard RoPE.
3. **StructEval-Bench v1 Construction** : Build 300–500 structurally-adversarial query-document pairs across Markdown, Tables, JSON. Define SDS (Structural Discrimination Score) and RIR metrics. Release on HuggingFace Datasets.
4. **Training + Ablation** : Train HiSE-Embed via contrastive learning with structurally-adversarial hard negatives. POMDP over structural state for tractable address inference at scale. Controlled ablation: same backbone · same data · same compute · only positional encoding changes.
5. **Evaluation + Release** :Validate on StructEval-Bench v1 (SDS ↑ significant) and BEIR/MTEB (nDCG@10 within ±2% margin). Release HiSE-Embed, StructEval-Bench v1, HiSA implementation under asl-research/hise.

Architecture change is the only variable. Every other factor held constant — this is what makes the claim provable.

System Architecture



Same text. Different structure. Different meaning- HiSE makes embedding models see the difference.

Modules & Functionalities

Four modules. One pipeline. Structure preserved end to end.

1. **Structural Parser & Addressing Engine:** Format-specific parsers extract structural coordinates from all input data types. Markdown AST walker extracts heading depth and path. Pandas extractor pulls row-column coordinates from tables. JSON recursive traverser extracts key-path depth and parent hash. HTML/XML DOM walker extracts node depth and sibling index. All parsers output a unified (m, d, p, s) tuple — same interface regardless of data format.
2. **HiSA Positional Encoding Layer:** Replaces standard RoPE in the transformer backbone. Accepts (m, d, p, s) tuple per token. Applies four independent learned rotation frequencies producing composite structural address $\theta = \theta_1 m + \theta_2 d + \theta_3 p + \theta_4 s$. Embedded directly into $Q \cdot K$ attention computation. 96 new scalar parameters total.
3. **POMDP Structural State Estimator:** Estimates composite structural address probabilistically when full document parse is computationally intractable. Maintains a belief state over structural position — depth, parent identity, sibling index — updating with each observed token. Processes just enough structure to achieve true-negative level precision without exhaustive tree computation at scale.
4. **Contrastive Training & Evaluation Pipeline:** Trains HiSE-Embed via InfoNCE loss with two negative categories: standard semantic hard negatives and structurally-adversarial hard negatives — identical surface text, structurally-opposed ground truth. Evaluates on StructEval-Bench v1 (SDS, RIR metrics) and BEIR/MTEB (nDCG@10) for general retrieval parity confirmation.

Each module is independently testable. Each has a defined input, output, and success criterion.

Technology & Requirements

- **Software Requirements:**

1. PyTorch 2.x · CUDA 12.8 · HuggingFace Transformers
2. Sentence-Transformers · custom HiSA module replacing RoPE
3. FAISS vector indexing · BEIR · MTEB · StructEval-Bench v1
4. Python-Markdown · pandas · JSON key-path parser
5. Weights & Biases · HuggingFace Hub for open-source release

- **Hardware Requirements:**

1. Cuda Core Processing systems · Unifiled Video Memory · primary training · ASL server
2. Local System Resources · Primary Memory · local development
3. High BandWidth system processor
4. 500GB SSD · checkpoints and embedding index storage
5. Random Access Memory · corpus loading and FAISS operations

Software:

All development and training containerized via Docker on ASL research infrastructure · reproducible builds guaranteed across environments.

Hardware:

Primary training on NVIDIA RTX (Blackwell architecture) · prototype validation on RTX · both locally and hosted dependency.

Expected outcome

Three artifacts. One architectural correction.

1. **HiSE-Embed (The Model) -** A contrastive retrieval embedding model under 300M parameters with HiSA replacing standard RoPE. Trained on structurally-adversarial hard negatives across tabular, JSON, HTML/XML, Markdown, and complex document formats. Released open-source on HuggingFace Hub under asl-research/hise-embed (Apache 2.0).
2. **StructEval-Bench v1 (The Benchmark)** - First retrieval benchmark explicitly designed to measure structural discrimination. 300–500 structurally-adversarial query-document pairs across Markdown heading hierarchies, tabular row-column structures, and nested JSON key-paths. Defines two new metrics — Structural Discrimination Score (SDS) and Retrieval Inversion Rate (RIR). Released on HuggingFace Datasets (CC BY 4.0).
3.

Metric	Target	Meaning
SDS on StructEval-Bench v1	↑ statistically significant ($p < 0.05$)	HiSA structurally discriminates
RIR on adversarial pairs	↓ measurable vs baseline	Wrong chunk no longer ranked first
nDCG@10 on BEIR/MTEB	≥ 0.46 within ±2% margin	No general retrieval regression
Parameter count	≤ 300M	Computationally accessible

Project Timeline

Phase	Timeline	Work	Success Criterion
Failure Survey	**Month 1–2**	Quantify structural discrimination gap on BGE-M3, Qwen3, Nomic across tabular, JSON, HTML, Markdown adversarial pairs	**Cosine sim ≥ 0.90 on structurally-opposed pairs. RIR baseline established**
HiSA Build	**Month 2-3**	Replace RoPE with 4D structural address. Build all format parsers outputting unified (m, d, p, s). Implement POMDP estimator	**Degenerate case: d=0, p=0, s=0 → identical output to standard RoPE**
Benchmark	**Month 3-4**	Build StructEval-Bench v1 — 300–500 adversarial pairs. Define SDS and RIR metrics. Release on HuggingFace Datasets	**Benchmark live with annotation guidelines**
Training	**Month 4-5**	Train HiSE-Embed with structurally-adversarial hard negatives. Run controlled ablation — only positional encoding changes	**SDS improvement statistically significant ($p < 0.05$) vs RoPE baseline**
Eval + Release	**Month 6**	Evaluate on BEIR/MTEB. Release HiSE-Embed, StructEval-Bench v1, HiSA code. Submit to ICNLP 2027	**nDCG@10 within ±2%. Submission by Nov 30, 2026**

Conclusion & Future Scope

Conclusion :

- HiSA — first positional encoding mechanism extending RoPE to 4D composite structural address across tabular, JSON, HTML/XML, Markdown and complex documents. POMDP makes it computationally tractable.
- HiSE-Embed — contrastive retrieval model under 300M params, structurally-adversarial hard negatives, format-agnostic, open-source on HuggingFace.
- StructEval-Bench v1 — first retrieval benchmark with structurally-adversarial pairs. Defines SDS and RIR — two metrics that did not exist before this work.
- Controlled ablation proving HiSA — not data or scale — drives structural retrieval improvement.
- StructRAG — production retrieval pipeline integrating HiSE-Embed into ASL DataSense.

Future Scope:

- HiSA → Generative LLMs — same multi Dim rotational mechanism applied to generative context windows. Structurally-aware generation, not just retrieval.
- Cross-lingual structural retrieval — extend HiSE to Indic languages where document hierarchy is the primary disambiguation signal.
- StructEval-Bench v2 — expand to legal contracts, clinical protocols, financial 10-K filings with real enterprise annotation.

HiSE does not improve an existing model. It corrects a broken assumption that position in a sequence is sufficient to represent position in data. It is not.

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Thank You !